

## ToDo

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# The `mousik` package\*

Celia Rubio Madrigal†

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## Abstract

This is a music-related package focused on writing music fragments. It works specially well with small examples, such as excerpts for theoretical articles. It is easier than other music writing methods in  $\text{\LaTeX}$  because it is based on the `tikz` package.

## Keywords

*mousik, music, music theory, music notation, music writing, score, sheet, tikz diagram, bar, clef, time signature, note, symbol, rest, dot, chord, accidental, sharp, flat, natural, key*

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\*This document corresponds to `mousik` v0.1, dated 2020/07/01.

†Email: celirubio.m@gmail.com

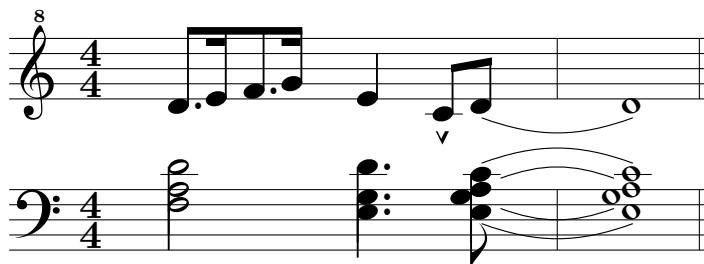
## 1 Introduction

There are already several ways to set music using L<sup>A</sup>T<sub>E</sub>X: `musixtex`, `lilypond` or `abc`, among others. Some of them are too difficult to use for not too complicated music; others do not work at all.

Based on the `tikz` package, the `mousik` package is perfect for short, one line music fragments. It may be used for longer scores, but there are more adaptable and lighter packages than this one. However, in spite of its compactness, the package is quite complete and versatile, but easy to learn and use.

## 2 The `mousik` environment

The way to draw a score is by means of the `mousik` environment. For longer examples, open the *examples* document.



```
\begin{mousik}
  \Clef{octave=8}
  \TimeSig{4}{4}

  \MDots{*/1, , */3, }
  \Note{|-,=|,|-,=|}{1,2,3,4}
  \Note{4}{2}
  \Artic{^}{0}
  \Note{8}{0,1}
  \Tie[e=2]{1}
  \Barline
  \Note{1}{1}

  \Barline[t=|.]
\end{mousik}
```

It has some options:

- `color` changes the background color. Default is white.
- `indent` indents the first line. Default is 0, in cm.
- `long` changes the maximum width of the score. Default is 100.

- `hsep` changes the horizontal separation. Default is 8.
- `vsep` changes the vertical separation in case of an overflow. Default is 2, in cm.
- `tikz` allows extra options for the outside `tikz` environment.

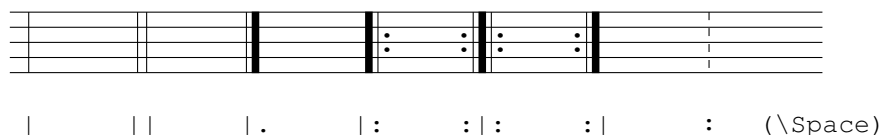
1.piano scores 2.single staff 3.resize 4.hsep with beams n dots 5.compatibility issues

## 3 Available commands

### 3.1 `\Barline`

The `\Barline` command prints a bar on the score. It has some options:

- `t` (type) is the type of bar. Default is `|`, the single bar.



- `xshift` moves the element horizontally (x axis). Default is 0, in cm.
- `yshift` moves the element vertically (y axis). Default is 0, in cm.

The only way to end a line in the score is with a bar or a space. Any other way may cause visual errors (see subsection 4.1).

### 3.2 `\Space`

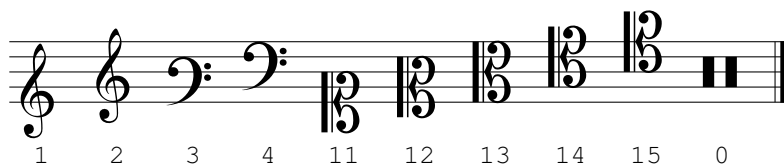
The `\Space` command is not only used when no bar is needed. It may also be used to input *extra* space between symbols. The default space is 1, in centimeters, —which is the default space between symbols in general, so it gets added— but it may be any number (see subsection 4.2).



### 3.3 `\Clef`

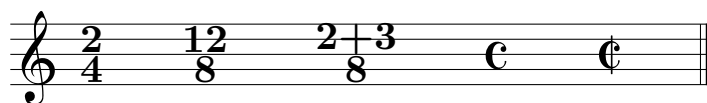
The `\Clef` command might be used at any moment on the score, although they are automatically printed if an overflow happens (see subsection 4.1). It has some options:

- `octave` prints an octave eight on the clef. If it is an 8, it prints it above the clef; if it is a -8, it prints it below the clef. Default is 0.
- `t` (type) is the type of clef. Default is 2 (the Treble Clef).
- `xshift` moves the element horizontally (x axis). Default is 0, in cm.
- `yshift` moves the element vertically (y axis). Default is 0, in cm.



### 3.4 `\TimeSig`

Time signatures might be used at any moment. `\TimeSig{2}{4}` will draw a 2 by 4 time signature. Any number or symbol can be written inside the brackets. Also `\TimeSig{C}{}` will produce a *common time* symbol and `\TimeSig{C|}{}` will produce an *alla breve* symbol.



It has some options:

- `xshift` moves the element horizontally (x axis). Default is 0, in cm.
- `yshift` moves the element vertically (y axis). Default is 0, in cm.

## 3.5 Notes and rests

### 3.5.1 `\Note`

The `\Note` command prints a note on the score. It accepts 2 compulsory arguments, as well as some options: **6.tuplet fucking sucks**

- `beam` chooses the width of the beam. Default is 1, in mm.
- `head` chooses the type of head. Default is . (round), but also x (cross) and o (hollow) are possible.
- `stem` chooses the height of the stem. Default is 0.5, in cm.
- `swap` prints the element upside down. Default is false, which means up if the height is less than 6, down otherwise.

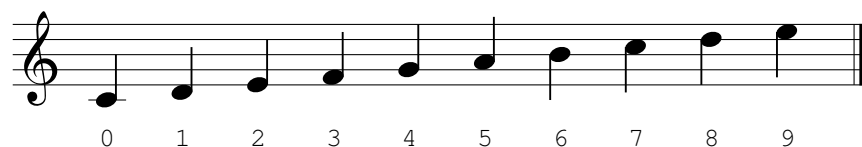
- *tuplet* prints the number of a tuplet. The written number is calculated by default, but it can be assigned optionally: `tuplet=5`.
- *xshift* moves the element horizontally (x axis). Default is 0, in cm.
- *yshift* moves the element vertically (y axis). Default is 0, in cm.

#### 7.stem in mm

The first compulsory argument is the duration of the note.

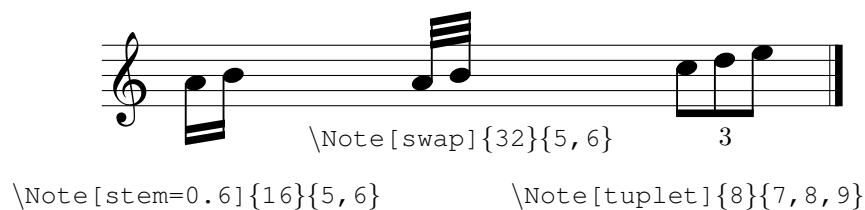


The second argument is the absolute height. Note that the scale is in milimeters. For example, these are the notes from 0 to 9:



The `\Note` command might also print several notes tied together with a beam, if the second argument is a list of heights separated by commas. Two notes will be tied together with an oblique beam, but three or more will have a straight beam.

#### 8.tuplet options: little bar, shifts



Notes with different durations that are tied together have a different syntax. Instead of using 8, 16 and 32 for the duration, each note has to be referenced with a -, a = or a ≡, respectively, and additional | might be needed for half the beam.

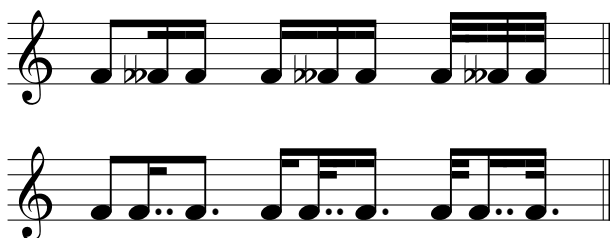


`\Note{|\equiv,-,|=,=,|=,-,|=,\equiv|}{0,1,2,3,4,3,2,1}`

To be able to write the  $\equiv$  (which is the Unicode character U+2261) as a symbol on the .tex file, I usually copy and paste it wherever I need it. Nonetheless, an alternative keyboard-friendly input that is equally valid is using `///` instead.

Additionally, spaces between notes from the same group might be useful —the default space is 0.5, in cm.

A space of 0.75 is obtained when a `!` is placed on both lists. A possible usage is when dealing with wide accidentals (see section 3.6.1) or double dots (see section 3.7.1).



A space of 1 is obtained when a space is placed on both lists. A possible usage is when dealing with rests (see section 3.5.2).



**9.fix beam =**

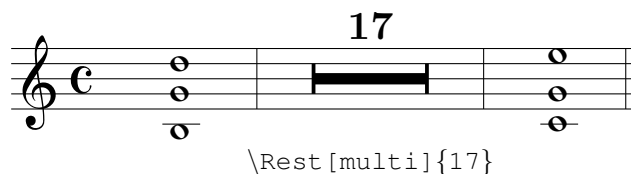
### 3.5.2 \Rest

The `\Rest` command prints a rest. It accepts its duration as an argument: 1, 2, 4, 8, 16 or 32.



It has some options:

- *multi* transforms the rest into a multimeasure rest —where the rest lasts an arbitrary amount of bars.

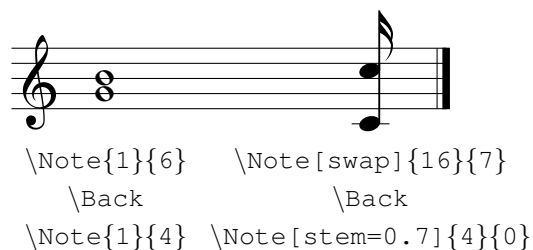


- *xshift* moves the element horizontally (x axis). Default is 0, in cm.
- *yshift* moves the element vertically (y axis). Default is 0, in cm.

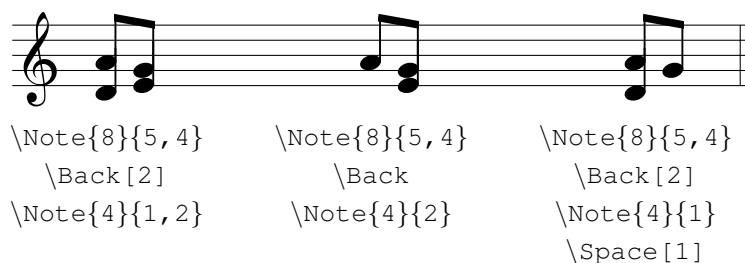
### 3.5.3 \Back

**10.maybe fix chord** Chords may be obtained by means of the `\Back` command. It accepts an optional argument: the number of notes to step backwards —mostly for the `\Note` command— with 1 as default.

The best way to use it is to call the `\Note` command with the highest notes; then call the `\Back` command; then call the `\Note` command with the duration 4 of a quarter note, to print them on top.



For extreme uses of the command, extra `\Space` commands might be needed.



## 3.6 Keys and accidentals

### 3.6.1 `\Acc`

The `Acc` command prints an accidental before a note. It accepts 2 arguments.

The first argument is the type of accidental: a number ranging from -2 to 2. The flat `b` symbol might also be called with a `b` and the sharp `#` symbol might be called with `\#`. **11.only # ?**

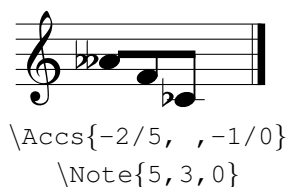


The second argument is the height of the symbol, which correspond to those of the notes.

It also has some options:

- `xshift` moves the element horizontally (x axis). Default is 0, in cm.
- `yshift` moves the element vertically (y axis). Default is 0, in cm.

When there are several notes, the `\Accs` command might be useful. It accepts as an argument a list of pairs: type/height. Where there is no accidental, the spot shall be left empty.



### 3.6.2 `\Key`

The `\Key` command prints the key at any moment on the score, although they are automatically printed if an overflow happens (see subsection 4.1).

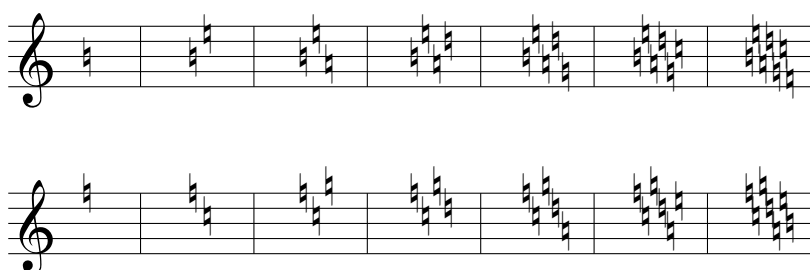
It accepts an argument: a positive whole number from 1 to 7 for sharps `#`, and a negative whole number from -1 to -7 for flats `b`.





### 3.6.3 \NoKey

**12.Mixed Keys** The `\NoKey` command is the same as the `\Key` command, but it prints naturals ♮ instead of sharps or flats.



Be careful: the `\Key` and `\NoKey` commands are affected by the current clef! Nonetheless, other commands such as `\Note` or `\Acc` are not affected and need absolute height values.



```
\begin{mousik}
  \Clef[t=4]
  \Key{1}
  \TimeSig{C|}{}

  \Acc{0}{8}
  \Note{1}{8}
  \Barline[t=|.]
\end{mousik}
```

## 3.7 Other symbols

### 3.7.1 \MDot

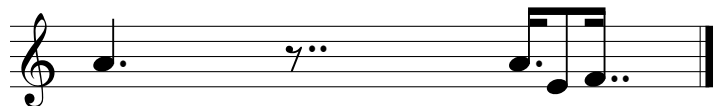
The `\MDot` command prints any amount of dots in front of a note or rest. It accepts a compulsory argument: the height of the dot, which corresponds to those of the notes.

It has some options:

- `t` (type) accepts `**` or `..` to produce 2 dots. The default is `*`.

- *xshift* moves the element horizontally (x axis). Default is 0, in cm.
- *yshift* moves the element vertically (y axis). Default is 0, in cm.

When there are several notes, the `\MDots` command might be useful. It accepts as an argument a list of pairs: symbols/height, where symbols might be `*` or `**`. Where there are no dots, the spot shall be left empty.



```

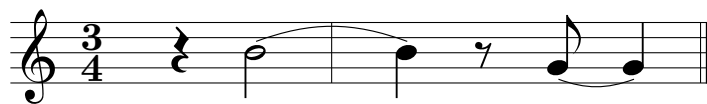
\MDot{5} \MDot[t= **]{7}\MDots{*/5, , **/3}
\Note{4}{5} \Rest{8} \Note{|=, -, =|}{5, 2, 3}

```

### 3.7.2 `\Tie`

The `\Tie` command draws a tie between two notes. It has a compulsory argument, the height of both notes, and some options:

- *angle* changes the angle in which it is drawn. Default is 20, in degrees.
- *b* (beginning) changes the starting point. Default is 0, in cm.
- *e* (end) changes the ending point. Default is 1, in cm.



```

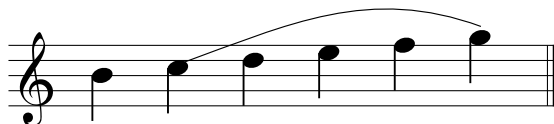
\Tie[e=2]{6} \Tie{4}

```

- *swap* prints the element upside down. Default is false, which means like a *u* if the height is less than 6, and like an *n* otherwise.
- *width* changes the line width. Default is 0.15, in mm.
- *xshift* moves the element horizontally (x axis). Default is 0, in cm.
- *yshift* moves the element vertically (y axis). Default is 0, in cm.

### 3.7.3 The **Slurred** environment

The `Slurred` environment draws a slur between the notes placed inside. It has two arguments: the heights of both beginning and end notes.



13.Fix angle

### 3.7.4 The Crescendo and Decrescendo environments

The Crescendo and Decrescendo environments draw a crescendo or decrescendo symbols, respectively, between the notes placed inside. It has an optional argument: the height of the symbol. The default value is 0.

14.Half line 15.obertura 16.shifts 17.weird spaces



### 3.7.5 \Artic and \Symbol

The \Artic command prints an articulation symbol on a successive note. It accepts two arguments. The first one is the type of articulation.

|               |         |         |         |      |
|---------------|---------|---------|---------|------|
| Staccatissimo | Marcato | Fermata | Mordent | Turn |
| {'}           | {^}     | {(.)}   | {~}     | {?}  |

|          |        |        |       |               |
|----------|--------|--------|-------|---------------|
| {.}      | {>}    | {-}    | {+}   | {~ }          |
| Staccato | Accent | Tenuto | Trill | Lower mordent |

|          |        |          |
|----------|--------|----------|
| Harmonic | Up bow | Down bow |
| {o}      | {v}    | {n}      |

The second one is the height of the symbol, which corresponds to those of the notes—but it might or might not be the height of the corresponding note.

It has some options:

- *xshift* moves the element horizontally (x axis). Default is 0, in cm.
- *yshift* moves the element vertically (y axis). Default is 0, in cm.

When there are several notes, the `\Artics` command might be useful. It accepts as an argument a list of pairs: symbols/height. Where there are no symbols, the spot shall be left empty.

The `\Symbol` command is similar to the `\Artic` command, but it only accepts the type of symbol and not the height. These are symbols that have a fixed position on the score.

|  |   |   |   |   |                |                   |
|--|---|---|---|---|----------------|-------------------|
|  | , | / | § | ∅ | Pedal<br>{Ped} | Lift Pedal<br>{*} |
|--|---|---|---|---|----------------|-------------------|



|        |         |       |        |      |      |   |
|--------|---------|-------|--------|------|------|---|
| {,}    | {/}     | {s\%} | {\%}   | {o+} | ped. | * |
| Breath | Caesura | Segno | Simile | Coda |      |   |

It also has some options:

- *xshift* moves the element horizontally (x axis). Default is 0, in cm.
- *yshift* moves the element vertically (y axis). Default is 0, in cm.

### 3.8 `\Text` and `\Dynamics`

The `\Text` command allows the printing of any text string. It has a compulsory argument: the text to be shown. It also has some options:

- *xshift* moves the element horizontally (x axis). Default is 0, in cm.
- *yshift* moves the element vertically (y axis). Default is 0, in cm.

To write anything in the usual font of dynamics, the `\Dynamics` command will work the same: `\Dynamics[yshift=-0.5]{pp}`. **18.dynamics wout yshift peta**

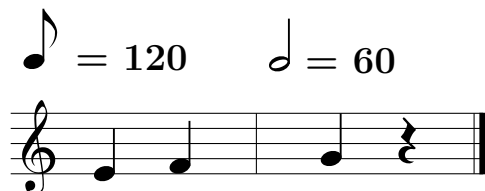
*rit.*



*pp*

### 3.9 `\Tempo`

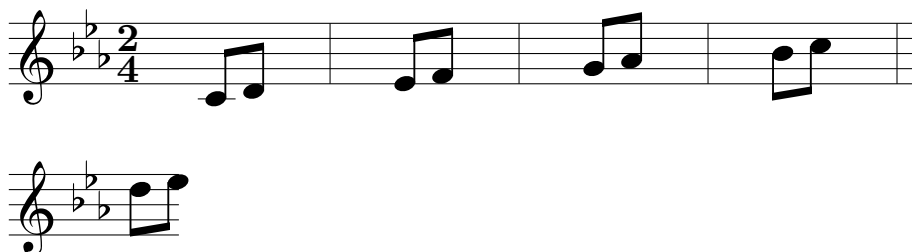
The `\Tempo` command prints a metronome tempo indicator. It has two arguments: the duration of the note and the numerical tempo which will be printed. It has the same options as the `\Text` command. An optional *yshift* of 2.5 is recommended.



## 4 Tips and observations

### 4.1 Overflow of lines

The `mousik` package is thought to be used for small fragments of music. The environment has a maximum length allowed—which might be changed with the *length* option—and whenever it is exceeded an overflow happens. The line is broken at that exact point and, if it does not coincide with a bar, something like this may happen:



The best way to avoid the error is by introducing extra spaces manually. The overflow will also automatically print the previous clef and key.

### 4.2 How do spaces work?

Spaces serve as the manual progression of an invisible cursor during the printing.

Most elements are separated by a distance of 1 cm, so a `\Space` command that has an argument smaller than -1 will actually step backwards. A `\Space` with a 0 as the argument does nothing.

The `\Space` command is truly relevant in this package. Not a lot of logic is internally implemented, so the user is supposed to adjust everything at their

convenience.

This is not to say that the package becomes too cumbersome to manage. For the general use I would say it is extremely user-friendly.

### 4.3 What if it does become too cumbersome: your own commands

If a concrete rhythm is constantly used on a music piece, but the commands become increasingly long and tedious, the user might want to summarize it by creating their own vocabulary.



```
\newcommand{\ttc}[2][ ]{\Note[#1]{|- , |=, =|}{#2}}
\newcommand{\tct}[2][ ]{\Note[#1]{|=, =|, -|}{#2}}

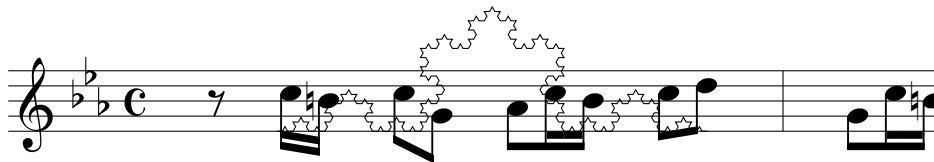
\begin{mousik}
  \Clef
  \ttc{7,8,9}
  \tct{10,9,8}
  \Barline[t= |.]
\end{mousik}
```

### 4.4 On your own: working with the tikz package

As stated in the introduction (section 1), the `mousik` package is based on the `tikz` package. In fact, the `mousik` environment is a disguised version of a `tikzpicture` environment, but with some variables and a printed staff.

This implies two separate things:

1. The `mousik` environment admits other commands inside, not just the commands explained in this text. For example, I might draw nodes, lines, or even a fractal behind my score:



2. The commands explained here might be used in a `tikzpicture` environment, and will appear with no staff. However, the command `\Init` should be used at the beginning to avoid initialization errors.

